

Stationary Action

One scalar function, one condition on it, and every equation of motion in this book.

WHERE WE ARE

Chapter 1.1 ended with a trick that should have unsettled you. To find the path a light ray takes through a stack of media, you did not track the ray. You wrote down a single number for each conceivable path — the total travel time — demanded that this number be *stationary*, and Snell's law fell out. A **global** statement about entire paths produced a **local** law about what happens at one interface.

That is a strange and powerful thing to be able to do, and 1.1 left it as an isolated curiosity about light. This chapter turns it into machinery. We build the branch of mathematics that differentiates a number with respect to a whole function — the **calculus of variations** — and then point it at mechanics.

Here is the destination, stated up front so you know what you are paying for. There is one scalar function L , called the Lagrangian. There is one quantity built from it, $S = \int L \, dt$, called the action. There is one condition, $\delta S = 0$. Out of that condition come Newton's laws (§4), the motion of anything with constraints (§6), Maxwell's equations (Chapter 2.6), Einstein's field equations (Chapter 3.6), the Dirac equation (Chapter 5.4), and the Standard Model (Chapter 6.7). Not analogues of them. Them.

This is a mathematics chapter. The physics in §4 onward is there to show you that the mathematics has teeth.

TOOLS YOU'LL NEED — Chapter 0.1: the derivative as the coefficient of the best linear approximation, and the $o(h)$ bookkeeping. This chapter is that definition with h replaced by an entire function; nothing else changes, which is the whole point of having defined it that way. Chapter 0.2: integration by parts — §3 is one application of it, and the boundary term matters. Chapter 0.6: the multivariable chain rule (§7 is nothing but), stationary points and the Hessian (§5), and Lagrange multipliers (§6.4, where the multiplier turns out to be a physical force). Chapter 1.1: Fermat's principle, and the catalogue of things the force picture cannot do.

1 · Functionals

A function eats a number and returns a number. A **functional** eats a *function* and returns a number. That is the entire definition, and the only reason it feels exotic is that you have spent your life with the first kind.

You have in fact been using functionals for years without the word. Here are three, and all three will be doing work before the chapter is out.

Arc length. Given a curve $y(x)$ between $x = a$ and $x = b$, its length is obtained by adding up little hypotenuses $ds = \sqrt{dx^2 + dy^2}$:

$$\mathcal{A}[y] = \int_a^b \sqrt{1 + y'(x)^2} dx. \quad (1.2.1)$$

Fermat's travel time. The same integral, weighted by the local slowness. If light moves at speed $c/n(x, y)$ where n is the refractive index, the time to traverse the curve is

$$\mathcal{T}[y] = \int_a^b \frac{n(x, y(x))}{c} \sqrt{1 + y'(x)^2} dx. \quad (1.2.2)$$

That is exactly the quantity Chapter 1.1 minimised, written out in general instead of for two flat slabs.

The action. Given a trajectory $x(t)$ between times t_1 and t_2 , and given some function $L(x, \dot{x}, t)$ of position, velocity and time,

$$S[x] = \int_{t_1}^{t_2} L(x(t), \dot{x}(t), t) dt. \quad (1.2.3)$$

We have not yet said what L is. That is deliberate; §5 is an honest discussion of where it comes from, and the answer is not what a textbook usually implies. For now L is any smooth function of three arguments, and S is the functional it defines.

Notice the notation: **square brackets**. $S[x]$ is not " S evaluated at the number x ". It is " S evaluated at the function x " — you must hand over the entire path, every $x(t)$ for every t in the interval, before S will give you a number back.

THE DOMAIN IS INFINITE-DIMENSIONAL, AND THAT IS THE WHOLE DIFFICULTY

To specify a point in $\mathbb{R}^3$ you supply three numbers. To specify a point in the domain of S you supply a value of x at *every one* of a continuum of times. The "variable" of the problem is a path, and the space of paths has infinitely many dimensions — one for each instant, loosely speaking.

So when we go looking for a stationary point of S , we are looking for a stationary point of a function on an infinite-dimensional space. Chapter 0.6 found stationary points of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ by setting n partial derivatives to zero. Here there is no finite list of partial derivatives to set to zero. We will get, instead, one condition *per instant of time* — which is to say a differential equation. That is the structural reason the answer to a variational problem is always a differential equation, and it is worth seeing before the algebra rather than after.

2 · The variational derivative

This section is the conceptual centre of the chapter. Everything after it is consequences.

We want to say what it means for the functional S to be stationary at a particular path. The honest difficulty is that we cannot write $\lim_{h \rightarrow 0} (S[x + h] - S[x])/h$, because the displacement is not a number h that we can divide by — it is an entire function. Dividing by a function is not a meaningful operation.

The way out is the one Chapter 0.1 insisted on, at some cost in apparent pedantry, on its first page. **The derivative is not a slope. It is the coefficient of the best linear approximation.** That definition never mentioned division, and it never mentioned one-dimensionality. It said: nudge the input, and identify the part of the output's response that is *linear in the nudge*. Nothing about that instruction breaks when the nudge is a function.

2.1 · Reduce to one variable

Fix the path $x(t)$ we want to test. Fix a second function $\eta(t)$ — the shape of the nudge — subject to one restriction:

$$\eta(t_1) = \eta(t_2) = 0. \quad (1.2.4)$$

Such an η is called an **admissible variation**. The restriction says the nudge does not move the endpoints: every path we compare starts at the same place at t_1 and finishes at the same place at t_2 . That is not a technicality — it is the boundary condition of the physical problem, and in §3 it is precisely what kills a term we would otherwise be stuck with.

Now build a one-parameter family of paths by scaling the nudge:

$$x_\epsilon(t) = x(t) + \epsilon \eta(t), \quad \epsilon \in \mathbb{R}. \quad (1.2.5)$$

At $\epsilon = 0$ this is the path we are testing; as ϵ grows the path bulges away from it in the shape η , always pinned at both ends. Feed the family into the functional:

$$\Phi(\epsilon) \equiv S[x + \epsilon \eta]. \quad (1.2.6)$$

Look carefully at what (1.2.6) is. Φ eats a single real number ϵ and returns a single real number. **It is an ordinary function of one real variable.** All of Chapter 0.1 applies to it verbatim, with no generalisation required at all. We have converted an infinite-dimensional problem into a one-dimensional one by agreeing, for the moment, to look in only one direction.

2.2 · The definition

Say that S is **stationary** at the path x if

$$\left. \frac{d}{d\epsilon} S[x + \epsilon \eta] \right|_{\epsilon=0} = 0 \quad \text{for every admissible } \eta. \quad (1.2.7)$$

The clause in *italics* is the entire content. Any single η probes one direction in the space of paths, and a function can easily be stationary along one direction and not others — a saddle is flat along two lines and steep everywhere else. Demanding (1.2.7) for *all* η is demanding flatness in every one of the infinitely many available directions at once. That is a very strong condition, and it is strong enough to determine the path.

The quantity in (1.2.7) gets a name and a symbol. Write

$$\delta S[x; \eta] \equiv \left. \frac{d}{d\epsilon} S[x + \epsilon\eta] \right|_{\epsilon=0}, \quad (1.2.8)$$

the **first variation** of S at x in the direction η . Unwinding (1.2.8) through Chapter 0.1's definition of a derivative, it is exactly the statement

$$S[x + \epsilon\eta] = \underbrace{S[x]}_{\text{value}} + \underbrace{\epsilon \delta S[x; \eta]}_{\text{linear in the nudge}} + \underbrace{o(\epsilon)}_{\text{dies faster}}. \quad (1.2.9)$$

THE CHEQUE WRITTEN ON PAGE ONE, CASHED HERE

Put (1.2.9) next to Chapter 0.1's central equation,

$$f(a + h) = f(a) + f'(a)h + o(h).$$

They are the same sentence. The dictionary is:

CHAPTER 0.1	HERE
the point a	the path $x(\cdot)$
the displacement h , a number	the variation $\eta(\cdot)$, a whole function
the derivative $f'(a)$, a number	the first variation $\delta S[x; \cdot]$, a linear <i>functional</i>
the linear term $f'(a)h$	the linear term $\delta S[x; \eta]$
error $o(h)$	error $o(\epsilon)$

Had we defined the derivative as "a slope", we would be stuck at the first row. There is no slope of a functional; there is no line to take the slope of, because the domain is a space of paths and there is no single direction to move in. But there is a best linear approximation, and it is unique, and it is what δS names.

Note what kind of object $\delta S[x; \cdot]$ is. It takes a function η and returns a number, and it does so *linearly*: $\delta S[x; \alpha\eta_1 + \beta\eta_2] = \alpha\delta S[x; \eta_1] + \beta\delta S[x; \eta_2]$, which you can read straight off (1.2.8) once §3 has computed it. A linear map whose target is $\mathbb{R}$ (Chapter 0.4) is what Chapter 0.6 called a **linear functional**, and, in finite dimensions, a **one-form** or covector. The derivative of a functional is a one-form on the space of paths — which is why it is written with a lower index in disguise, as the *coefficient* against which you integrate. That sentence will be true word for word in Chapter 5.5, with the path replaced by a field configuration.

2.3 · The functional derivative

One more piece of notation, because physics uses it constantly and it is easy to be mystified by it. In finite dimensions, Chapter 0.6 wrote the derivative of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ in the direction $\mathbf{h}$ as a sum over components,

$$df(\mathbf{h}) = \sum_{i=1}^n \frac{\partial f}{\partial x^i} h^i. \quad (1.2.10)$$

The gradient components $\partial f / \partial x^i$ are the *coefficients* of the linear map, one per direction. Here the directions are labelled not by an index i running over $1, \dots, n$ but by a time t running over $[t_1, t_2]$, and the sum over the label

becomes an integral over it. So we define the **functional derivative** $\delta S/\delta x(t)$ to be the coefficient function appearing in

$$\delta S[x; \eta] = \int_{t_1}^{t_2} \frac{\delta S}{\delta x(t)} \eta(t) dt. \quad (1.2.11)$$

Compare (1.2.11) with (1.2.10): index $\rightarrow$ time, sum $\rightarrow$ integral, gradient component $\rightarrow$ functional derivative. It is a continuum gradient. Section 3 is going to compute it, and the answer will be the Euler–Lagrange expression; the condition $\delta S = 0$ for all η will then become the statement that this continuum gradient vanishes at every t , which is one equation per instant, as promised.

3 · Euler–Lagrange, derived

Now do the calculation. Take $S[x] = \int_{t_1}^{t_2} L(x, \dot{x}, t) dt$ with L smooth, and compute δS . There are exactly four moves, and you have all four already.

3.1 · Move one: differentiate under the integral sign

Write out $\Phi(\epsilon)$. Since $x_\epsilon = x + \epsilon\eta$ implies $\dot{x}_\epsilon = \dot{x} + \epsilon\dot{\eta}$,

$$\Phi(\epsilon) = \int_{t_1}^{t_2} L(x + \epsilon\eta, \dot{x} + \epsilon\dot{\eta}, t) dt. \quad (1.2.12)$$

We want $\Phi'(0)$. The integrand depends on ϵ ; the limits do not. Assume for the moment that we may move the ϵ -derivative inside:

$$\frac{d\Phi}{d\epsilon} = \int_{t_1}^{t_2} \frac{\partial}{\partial \epsilon} L(x + \epsilon\eta, \dot{x} + \epsilon\dot{\eta}, t) dt. \quad (1.2.13)$$

► **Quoted.** The interchange in (1.2.13) is the one step in this derivation we are not proving. The standard sufficient condition is: if L and its ϵ -partial derivative are continuous on the rectangle $[t_1, t_2] \times [-\epsilon_0, \epsilon_0]$, then Φ is differentiable and (1.2.13) holds. Since we are assuming L is a smooth function of its three arguments and x, η are smooth on a closed bounded interval, that condition is met. The grind box below says what can go wrong when it is not.

3.2 · Move two: linearize the integrand

The ϵ -derivative inside is the multivariable chain rule of Chapter 0.6, applied to L as a function of its first two slots. Write $\partial L/\partial x$ and $\partial L/\partial \dot{x}$ for the partial derivatives of L with respect to its first and second arguments. Then, since the first argument moves at rate η and the second at rate $\dot{\eta}$ as ϵ increases,

$$\frac{\partial}{\partial \epsilon} L(x + \epsilon \eta, \dot{x} + \epsilon \dot{\eta}, t) = \frac{\partial L}{\partial x} \eta + \frac{\partial L}{\partial \dot{x}} \dot{\eta}, \quad (1.2.14)$$

where after setting $\epsilon = 0$ the partials are evaluated on the original path $(x(t), \dot{x}(t), t)$. So

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial x} \eta(t) + \frac{\partial L}{\partial \dot{x}} \dot{\eta}(t) \right] dt. \quad (1.2.15)$$

We are stuck, and it is worth being precise about why. To conclude anything about the integrand we need η to appear as an overall factor, so that "this holds for every η " has bite. It does in the first term. In the second term what appears is $\dot{\eta}$, which is *not* independent of η — it is determined by it. We cannot choose them separately, so we cannot argue term by term.

3.3 · Move three: integrate by parts

This is exactly the situation Chapter 0.2 built integration by parts for: *you may move a derivative from one factor onto the other, at the cost of a minus sign and a boundary term*. Apply it to the second term with $u = \partial L / \partial \dot{x}$ and $v' = \dot{\eta}$, so $v = \eta$:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}} \dot{\eta} dt = \left[\frac{\partial L}{\partial \dot{x}} \eta \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) \eta dt. \quad (1.2.16)$$

Now look at the boundary term. It is $(\partial L / \partial \dot{x})|_{t_2} \eta(t_2) - (\partial L / \partial \dot{x})|_{t_1} \eta(t_1)$, and by (1.2.4) — the requirement that admissible variations vanish at the endpoints — *both* η 's are zero. The boundary term dies.

It dies **because we fixed the endpoints**, not because of any property of L . That is worth remembering rather than deriving twice, because when you eventually stop fixing the endpoints — which happens in Chapter 1.3, in Chapter 3.6 for gravity, and everywhere in field theory where the "boundary" is at spatial infinity — the surviving boundary term is not litter. It is where the momentum lives, and where the black-hole entropy will come from.

Substituting (1.2.16) into (1.2.15) and collecting the common factor η :

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) \right] \eta(t) dt. \quad (1.2.17)$$

Compare with (1.2.11) and read off the answer to §2.3: the functional derivative of the action is the bracket,

$$\frac{\delta S}{\delta x(t)} = \frac{\partial L}{\partial x} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right). \quad (1.2.18)$$

3.4 · Move four: the fundamental lemma

Stationarity says (1.2.17) vanishes for every admissible η . We want to conclude that the bracket vanishes. That step is not free — it is a genuine theorem, and skipping it is skipping the move that converts "the integral is zero" into "the

integrand is zero". The same move recurs in Chapter 5.2 to produce field equations and in Chapter 3.6 to produce Einstein's, so we prove it once, here, properly.

FUNDAMENTAL LEMMA OF THE CALCULUS OF VARIATIONS

Let f be continuous on $[t_1, t_2]$. If

$$\int_{t_1}^{t_2} f(t) \eta(t) dt = 0$$

for every continuously differentiable η with $\eta(t_1) = \eta(t_2) = 0$, then $f(t) = 0$ for all $t \in [t_1, t_2]$.

The theorem is not obvious and it is not trivially true. An integral can vanish through cancellation — $\int_{-1}^1 t dt = 0$ without t being zero anywhere but the origin. The hypothesis defeats cancellation by allowing you to *choose* η : if f is positive somewhere, pick an η that is supported only there, and there is nothing left to cancel against. Making that idea airtight requires exhibiting such an η , which is the content of the proof.

Proof. Suppose, for contradiction, that f is not identically zero. Then there is a point t_0 with $f(t_0) \neq 0$; replacing f by $-f$ if necessary (which changes nothing, since the hypothesis is preserved), assume $f(t_0) > 0$. If t_0 is an endpoint, continuity gives a nearby interior point where f is still positive, so assume t_0 is interior.

Set $c = \frac{1}{2}f(t_0) > 0$. By continuity of f at t_0 , there is a $\delta > 0$ with $[t_0 - \delta, t_0 + \delta] \subset (t_1, t_2)$ and

$$f(t) > c \quad \text{for all } t \text{ with } |t - t_0| < \delta. \quad (1.2.19)$$

Now construct the probe. Define

$$\eta(t) = \begin{cases} [\delta^2 - (t - t_0)^2]^2, & |t - t_0| \leq \delta, \\ 0, & \text{otherwise.} \end{cases} \quad (1.2.20)$$

Three things about (1.2.20), each checkable by hand. It is *non-negative everywhere*, and *strictly positive* on the open interval $|t - t_0| < \delta$. It *vanishes at t_1 and t_2* , since it vanishes outside a subinterval strictly inside $[t_1, t_2]$. And it is *continuously differentiable*: away from the two junction points this is clear, and at $t = t_0 \pm \delta$ both the value $[\delta^2 - (t - t_0)^2]^2$ and its derivative $-4(t - t_0)[\delta^2 - (t - t_0)^2]$ approach zero from inside, matching the zero function outside. So η is admissible.

Feed it in. The integrand vanishes outside the bump, and inside it both factors are positive:

$$\int_{t_1}^{t_2} f \eta dt = \int_{t_0 - \delta}^{t_0 + \delta} f \eta dt > c \int_{t_0 - \delta}^{t_0 + \delta} \eta dt = c \cdot \frac{16 \delta^5}{15} > 0. \quad (1.2.21)$$

That contradicts the hypothesis that the integral vanishes for every admissible η . Hence no such t_0 exists and $f \equiv 0$.

■

The number $16\delta^5/15$. Substitute $u = t - t_0$ and expand:

$$\int_{-\delta}^{\delta} (\delta^2 - u^2)^2 du = \int_{-\delta}^{\delta} (\delta^4 - 2\delta^2 u^2 + u^4) du = 2 \left(\delta^5 - \frac{2\delta^5}{3} + \frac{\delta^5}{5} \right) = \frac{16\delta^5}{15}.$$

The value is irrelevant to the proof — all that matters is that it is strictly positive — but it is worth computing once so that the inequality in (1.2.21) is not taken on faith.

A smoother bump. Our η is C^1 but not C^2 : differentiate twice and at $t = t_0 \pm \delta$ you get $8\delta^2$ from inside and 0 from outside. C^1 is all the lemma needs, because the Lagrangian depends on x and $\dot{x}$ and no higher derivative. If you want a variation that is infinitely differentiable — and you will, in field theory, where Lagrangians with second derivatives occur — use

$$\eta(t) = \exp\left(\frac{-1}{\delta^2 - (t - t_0)^2}\right) \quad \text{for } |t - t_0| < \delta, \quad \eta = 0 \text{ otherwise.}$$

Every derivative of this vanishes at the junctions, because $e^{-1/s}$ and all its derivatives tend to zero as $s \rightarrow 0^+$ faster than any power (Chapter 0.3's remark that this function has a Taylor series that is identically zero and yet is not zero — the standard counterexample to "smooth implies analytic").

Restricting to C^∞ probes only *strengthens* the hypothesis of the lemma, so the conclusion still follows; you simply have fewer probes and still enough.

Continuity of f is essential. Drop it and the lemma is false in the pointwise sense: a function that is 1 at a single point and 0 elsewhere has $\int f\eta = 0$ for every η , yet is not identically zero. The correct statement without continuity is " $f = 0$ almost everywhere". Physically this is the statement that the equation of motion holds at every instant *except possibly on a set of measure zero* — which is exactly the loophole that permits collisions, shocks, and the kinked worldlines of Chapter 5.8's path integral.

The interchange, and when it fails. The condition quoted in §3.1 is sufficient, not necessary, and it is a genuine condition. The standard failure is $\Phi(\epsilon) = \int_0^1 \frac{\epsilon}{\epsilon^2 + t^2} dt = \arctan(1/\epsilon)$ for $\epsilon \neq 0$, which tends to $\pi/2$ as $\epsilon \rightarrow 0^+$ and to $-\pi/2$ from below, while the pointwise limit of the integrand is 0 for every $t \neq 0$. The integrand's ϵ -derivative is not continuous at $(\epsilon, t) = (0, 0)$, and the interchange is invalid. In mechanics, with L smooth and the interval compact, this never bites; in field theory with fields that do not decay it sometimes does, and the symptom is a divergent action.

3.5 · The Euler–Lagrange equation

Apply the lemma to (1.2.17). The bracket is continuous (we assumed L smooth and the path smooth), and the integral vanishes for every admissible η . Therefore the bracket vanishes identically, and we have

$$\boxed{\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} = 0} \quad (1.2.22)$$

for all $t \in [t_1, t_2]$. This is the **Euler-Lagrange equation**. It is a second-order ordinary differential equation for $x(t)$ — second order because $\partial L / \partial \dot{x}$ depends on $\dot{x}$, and differentiating it in time produces $\ddot{x}$. Two constants of integration, fixed by the two endpoint conditions. Everything is consistent.

Nothing in the derivation used the fact that there was one coordinate. Suppose the configuration is described by n numbers $q^1, \dots, q^n$ and $L = L(q^1, \dots, q^n, \dot{q}^1, \dots, \dot{q}^n, t)$. Vary each independently: take $\eta = (\eta^1, \dots, \eta^n)$ and run §3.1–§3.4 with sums over i throughout. Move two produces $\sum_i [\partial_{\dot{q}^i} L \eta^i + \partial_{\ddot{q}^i} L \dot{\eta}^i]$, move three integrates each term by parts separately, and move four is applied n times — once with η^1 arbitrary and all other $\eta^i = 0$, once with η^2 arbitrary, and so on. The result is

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = 0, \quad i = 1, \dots, n. \quad (1.2.23)$$

One equation per degree of freedom, all of them from the single scalar L . That compression — n coupled second-order differential equations packaged in one function — is the first practical reason to care, and by §6 it will look like a bargain.

▼ Grind box — what happens if the Lagrangian contains $\ddot{q}$

Nothing in §3 required L to stop at first derivatives. Suppose $S = \int_{t_1}^{t_2} F(q, \dot{q}, \ddot{q}, t) dt$. Moves one and two are unchanged and give

$$\delta S = \int_{t_1}^{t_2} \left[\frac{\partial F}{\partial q} \eta + \frac{\partial F}{\partial \dot{q}} \dot{\eta} + \frac{\partial F}{\partial \ddot{q}} \ddot{\eta} \right] dt.$$

Move three now has to be run *twice* on the last term:

$$\begin{aligned} \int \frac{\partial F}{\partial \ddot{q}} \ddot{\eta} dt &= \left[\frac{\partial F}{\partial \ddot{q}} \dot{\eta} \right]_{t_1}^{t_2} - \int \frac{d}{dt} \left(\frac{\partial F}{\partial \ddot{q}} \right) \dot{\eta} dt \\ &= \left[\frac{\partial F}{\partial \ddot{q}} \dot{\eta} - \frac{d}{dt} \left(\frac{\partial F}{\partial \ddot{q}} \right) \eta \right]_{t_1}^{t_2} + \int \frac{d^2}{dt^2} \left(\frac{\partial F}{\partial \ddot{q}} \right) \eta dt. \end{aligned}$$

There are now *two* boundary terms, and killing them needs both η and $\dot{\eta}$ to vanish at the ends — which is to say the variational problem is only well posed if you fix the position *and* the velocity at each endpoint. Given that, the fundamental lemma applies verbatim and delivers

$$\frac{\partial F}{\partial q} - \frac{d}{dt} \left(\frac{\partial F}{\partial \dot{q}} \right) + \frac{d^2}{dt^2} \left(\frac{\partial F}{\partial \ddot{q}} \right) = 0,$$

a *fourth-order* equation, and the obvious pattern continues with alternating signs for still higher derivatives. This is the version the Familiar Ground callout in §7 needs: with $F = \lambda f''(x)^2$ the

equation reduces to $f'''' = 0$, whose solutions are the cubics.

■ One quoted warning, because it explains an omission you will otherwise notice. Ostrogradsky proved that a Lagrangian genuinely depending on $\ddot{q}$ has a conserved energy that is *linear* in one of its momenta and therefore unbounded below — such theories are generically unstable. That is why every fundamental action in §8.1's table contains at most first derivatives of its fields. The Einstein–Hilbert action looks like an exception, since R contains second derivatives of the metric; it escapes because those second derivatives enter only inside a total divergence, as Chapter 3.6 will show explicitly.

4 · It reproduces Newton

Time to check that this machinery is not describing some other universe. Take a single particle of mass m moving in one dimension in a potential $V(x)$, and make the choice

$$L = T - V = \frac{1}{2}m\dot{x}^2 - V(x). \quad (1.2.24)$$

Where that choice comes from is §5's problem. Right now we only want its consequences. Compute the two partial derivatives, treating x and $\dot{x}$ as independent slots of the function L — which is legitimate for the reason spelled out in the warning callout below:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial L}{\partial x} = -\frac{dV}{dx}. \quad (1.2.25)$$

Substitute into (1.2.22):

$$\frac{d}{dt}(m\dot{x}) + \frac{dV}{dx} = 0 \quad \Longleftrightarrow \quad m\ddot{x} = -\frac{dV}{dx}. \quad (1.2.26)$$

The right-hand side is the force, by Chapter 1.1's definition of a conservative force as minus the gradient of its potential. So (1.2.26) is $m\ddot{x} = F$.

In three dimensions nothing changes except the number of equations. With $L = \frac{1}{2}m\dot{\mathbf{x}} \cdot \dot{\mathbf{x}} - V(\mathbf{x})$ and $q^i = x^i$,

$$\frac{\partial L}{\partial \dot{x}^i} = m\dot{x}^i, \quad \frac{\partial L}{\partial x^i} = -\frac{\partial V}{\partial x^i}, \quad \Longrightarrow \quad m\ddot{x}^i = -\frac{\partial V}{\partial x^i} = F^i. \quad (1.2.27)$$

Now read (1.2.27) again, slowly, without substituting anything. The object $\partial L / \partial \dot{x}^i$ turned out to be $m\dot{x}^i$ — the **momentum**. The object $\partial L / \partial x^i$ turned out to be $-\partial V / \partial x^i$ — the **force**. So the Euler–Lagrange equation, in Cartesian coordinates with $L = T - V$, reads

$$\frac{dp_i}{dt} = F_i. \quad (1.2.28)$$

Newton's second law is not a separate law that happens to agree with the action principle. It is the Euler–Lagrange equation, written in Cartesian coordinates. That is the entire relationship between Chapter 1.1 and this one.

This also explains why the notation generalises the way it does. For an arbitrary Lagrangian and arbitrary coordinates we *define*

$$p_i \equiv \frac{\partial L}{\partial \dot{q}^i} \quad (\text{the canonical momentum conjugate to } q^i), \quad (1.2.29)$$

whereupon (1.2.23) always reads $\dot{p}_i = \partial L / \partial q^i$. The definition is not a whim: it is chosen so that the Euler–Lagrange equation keeps the shape of Newton's law. Chapter 1.3 takes p_i seriously as an independent variable and gets Hamiltonian mechanics; Chapter 1.4 shows that whenever L fails to depend on some q^i at all, the corresponding p_i is conserved — which is Noether's theorem in its simplest guise, already visible in (1.2.23) for anyone who looks.

△ WHY THIS ISN'T OBVIOUS

In (1.2.25) we differentiated L with respect to $\dot{x}$ while holding x fixed. That should bother you. Along an actual trajectory $\dot{x}$ is not free to vary independently of x — it is x 's own derivative. Holding one fixed while changing the other looks like nonsense.

The resolution is to be careful about what L is a function of. L is a function of three independent arguments — call them (u, v, t) — defined on a space of its own, with no relationship between u and v whatsoever. For the free particle, $L(u, v, t) = \frac{1}{2}mv^2 - V(u)$ is defined for every pair $(u, v) \in \mathbb{R}^2$, and $\partial L / \partial v = mv$ is a perfectly ordinary partial derivative of that perfectly ordinary function of two variables. The constraint " v is the time derivative of u " is imposed *only when you evaluate*: the action (1.2.3) substitutes $u = x(t)$ and $v = \dot{x}(t)$ into L *after* the partial derivatives have been taken.

The precise statement is that L is a function on the **tangent bundle** of the configuration space — the space whose points are (position, velocity) pairs, with both free. A path in configuration space lifts to a curve in the tangent bundle by attaching to each point its own velocity, and that lift is where the constraint lives. Chapter 3.2 constructs the tangent bundle properly; for now the operational rule is exact and safe: **differentiate L symbolically, treating x and $\dot{x}$ as unrelated letters; only afterwards put the path in.** The $\frac{d}{dt}$ in (1.2.22) is the one operation that does know about the path — it is a total time derivative along the trajectory, and it is why $\frac{d}{dt}(\partial L / \partial \dot{x})$ generates $\ddot{x}$ terms while $\partial L / \partial \dot{x}$ alone does not.

5 · Why $T - V$? An honest answer

You have just watched $L = T - V$ reproduce Newtonian mechanics exactly. The natural next question is where $T - V$ came from, and the honest answer has three parts, of which only the third is satisfying.

Part one: within classical mechanics, it is not derived. There is no argument that starts from Newton's laws and ends at $L = T - V$ by necessity. What there is, is the calculation of §4 run backwards: if you want a variational principle whose Euler–Lagrange equation is $m\ddot{x} = -V'(x)$, then $\frac{1}{2}m\dot{x}^2 - V(x)$ is a Lagrangian that does the job. It is not the only one — §"Your turn" problem 4 shows you may add any total time derivative without changing a thing, and multiplying L by a nonzero constant also leaves the equations alone. $T - V$ is a *choice*, justified after the fact by the fact that it works. Anyone who tells you the minus sign is intuitive is selling something.

Part two: "least action" is the wrong word. The condition we imposed, (1.2.7), is that the first variation vanishes. That is *stationarity*, exactly as in Chapter 0.6 — and Chapter 0.6 was emphatic that a vanishing gradient signals a maximum, a minimum, or a saddle, and that in more than one dimension saddles are the generic case. The space of paths has infinitely many dimensions. Saddles are not the exception there; they are ordinary. The warning callout below makes this concrete with a system where the action is provably not a minimum.

Part three: the real reason, which is quantum. In Chapter 5.8 we will construct the Feynman path integral, in which a quantum particle going from A to B contributes an amplitude from *every* path, each weighted by the phase

$$e^{iS[x]/\hbar}. \tag{1.2.30}$$

The total amplitude is the sum over all paths. Now consider what that sum does. Away from a stationary point of S , neighbouring paths have actions differing at *first* order in the separation, so their phases differ at first order too; since $\hbar$ is tiny, $S/\hbar$ swings through many multiples of 2π over a microscopic change of path, and the contributions from neighbouring paths point in every direction on the complex plane and cancel. Near a stationary point, by construction, S changes only at *second* order — a whole neighbourhood of paths shares essentially the same phase, and their contributions add rather than cancel.

So the classical path is not selected because nature computes an integral and minimises it. **Nature does not extremise anything. The classical path is simply where the quantum amplitudes fail to cancel.** The action principle is a statement about constructive interference that survives into the $\hbar \rightarrow 0$ limit, and the reason the classical path is the stationary one is that stationary phase is the only place a wildly oscillating sum can leave a residue. The reason $L = T - V$ and not something else is then pushed back one level: it is whatever sits in the exponent of the quantum theory, and *that* gets fixed in Chapter 5.3 by symmetry requirements — Lorentz invariance, locality, and the demand that the theory be unitary.

▀ That paragraph is a **promise**, not a derivation. Nothing in it is proved here; it is stated so that you know a real explanation exists and where it is, and so that you do not spend the next six chapters believing that "nature is economical" is an explanation. It isn't. Chapter 5.8 redeems the promise, including the stationary-phase approximation that makes "wildly oscillating integrals localise on stationary points" a theorem rather than a story.

5.1 · Seeing stationarity

Before the abstractions pile up, look at the thing itself. The figure below takes the simplest system with a curved trajectory — a particle thrown straight up in uniform gravity — and lets you deform its path while watching the

action.

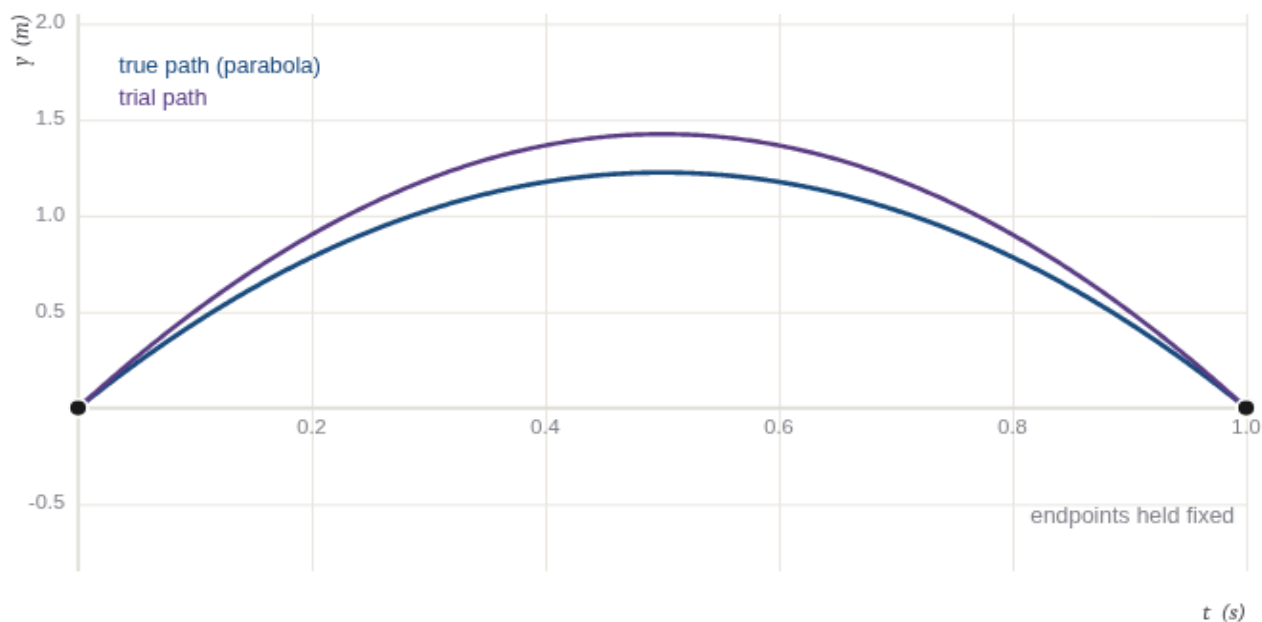
The setup: mass $m = 1$ kg, gravity $g = 9.8 \text{ m s}^{-2}$, and a path that starts at $y = 0$ at $t = 0$ and returns to $y = 0$ at $t = T = 1$ s. The Euler–Lagrange equation (1.2.26) with $V = mgy$ gives $\ddot{y} = -g$, whose solution with those endpoints is the parabola $y_*(t) = \frac{1}{2}gt(T - t)$, peaking at 1.225 m. Its action, worked out directly,

$$\begin{aligned} S[y_*] &= \int_0^T \left(\frac{1}{2}m\dot{y}_*^2 - mgy_* \right) dt = \frac{mg^2T^3}{24} - \frac{mg^2T^3}{12} = -\frac{mg^2T^3}{24} \\ &= -4.0017 \text{ J s}. \end{aligned} \tag{1.2.31}$$

The trial path is $y(t) = y_*(t) + \eta(t)$ with

$$\eta(t) = a_1 \sin \frac{\pi t}{T} + a_2 \sin \frac{2\pi t}{T} + a_3 \sin \frac{3\pi t}{T}, \tag{1.2.32}$$

which vanishes at $t = 0$ and $t = T$ for any a_k , so every trial path is admissible. The three sliders are the a_k . The widget computes $\int T dt$ and $\int V dt$ by numerical quadrature along whatever path is on screen — sampling the curve, differencing it to get the velocity, and summing — so the numbers are measurements of the drawn path, not a formula.



a_1 a_2
 a_3

halve the deformation

snap to true path

integrand: $T - V$

$\int T \, dt = 5.3481 \, \text{J s}$ $\int V \, dt = 9.2511 \, \text{J s}$

$\int (T - V) \, dt$: trial -3.9030 | true path -4.0017

Δ from true path = +0.098696

press "halve" twice to watch ΔS fall by a factor of four each time

Stationary, and only for $T - V$. Blue is the true parabola; purple is your trial path. **(i) Stationarity.** Move any slider a little and watch: the purple curve visibly detaches from the blue one while the action barely moves. At $a_1 = 0.1 \, \text{m}$ — a bulge you can see plainly, 8% of the peak height — the action changes by $+0.0247$ out of -4.0017 , six parts in a thousand. That is what "stationary" looks like: first-order motion of the path, second-order response of the number. **(ii) Second order, measured.** Press *halve* repeatedly. The readout reports the ratio of the new ΔS to the old one, and it is 0.2500 every time — the signature of a quantity that starts at ϵ^2 . **(iii) The minus sign is not free.** Press the toggle to integrate $T + V$ instead. The physical path is now nowhere near stationary: nudging a_1 by ± 0.01 moves the number by about ± 0.125 , five hundred times the response of $T - V$ to the same nudge, and the response reverses sign with the nudge, which is the unmistakable signature of a non-vanishing first derivative. $T - V$ is not a convention. It is the combination for which this particular path is a stationary point, and no other combination of T and V with constant coefficients will do it.

Two facts the figure is showing can be established exactly, and it is worth doing so rather than trusting the pixels. Write the trial path as $y_* + \eta$ with η vanishing at both ends. Because L is quadratic in $\dot{y}$ and linear in y , the expansion of S terminates:

$$\begin{aligned}
S[y_* + \eta] &= \int_0^T \left[\frac{1}{2} m (\dot{y}_* + \dot{\eta})^2 - mg(y_* + \eta) \right] dt \\
&= S[y_*] + \int_0^T (m\dot{y}_*\dot{\eta} - mg\eta) dt + \int_0^T \frac{1}{2} m \dot{\eta}^2 dt
\end{aligned} \tag{1.2.33}$$

with *no* higher terms. Integrate the middle bracket's first piece by parts: $\int m\dot{y}_*\dot{\eta} dt = [m\dot{y}_*\eta]_0^T - \int m\ddot{y}_*\eta dt$. The boundary term vanishes because η does; and $\ddot{y}_* = -g$, so the remaining integral is $+\int mg\eta dt$, which cancels the second piece exactly. The first variation is zero — that is stationarity, verified by hand. What survives is

$$\Delta S = S[y_* + \eta] - S[y_*] = \int_0^T \frac{1}{2} m \dot{\eta}^2 dt \geq 0, \tag{1.2.34}$$

manifestly non-negative and zero only for $\eta \equiv 0$. For this system the action is a genuine *minimum*, not merely stationary. And with the specific η of (1.2.32), the sine modes are orthogonal on $[0, T]$ (Chapter 0.9), so the cross terms integrate to zero and

$$\Delta S = \frac{m\pi^2}{4T} (a_1^2 + 4a_2^2 + 9a_3^2). \tag{1.2.35}$$

Purely quadratic in the amplitudes, which is why halving them quarters ΔS . At $a_1 = 0.1, a_2 = a_3 = 0$ this predicts $\Delta S = \pi^2(0.01)/4 = 0.024674$, and the widget's quadrature reports 0.024674. The number on the screen is not decorative.

Run the same expansion on $\tilde{S} = \int (T + V) dt$ and the cancellation fails: the sign flip turns $-\int mg\eta$ into $+\int mg\eta$, so instead of cancelling the by-parts term it doubles it, leaving a first variation $\delta\tilde{S} = 2mg \int_0^T \eta dt$. For the fundamental mode that is $2mgT \cdot (2/\pi)a_1 = 4mgTa_1/\pi$, a slope of 12.48 per metre of amplitude — which is the ± 0.125 the caption quotes for $a_1 = \pm 0.01$. Not stationary, and not by a little.

▲ WHY THIS ISN'T OBVIOUS

The principle is called "least action" and the quantity is very often not least. Here is a system where you can prove it in four lines.

Take the harmonic oscillator, $L = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}m\omega^2x^2$, and the path $x_*(t) \equiv 0$ on $[0, \mathcal{T}]$, which satisfies the Euler–Lagrange equation $\ddot{x} = -\omega^2x$ and both endpoint conditions $x(0) = x(\mathcal{T}) = 0$. Since L is exactly quadratic, the expansion again terminates, and since the first variation vanishes,

$$\Delta S = \frac{m}{2} \int_0^{\mathcal{T}} (\dot{\eta}^2 - \omega^2 \eta^2) dt.$$

Choose the admissible variation $\eta = A \sin(\pi t/\mathcal{T})$. Using $\int_0^{\mathcal{T}} \sin^2(\pi t/\mathcal{T}) dt = \int_0^{\mathcal{T}} \cos^2(\pi t/\mathcal{T}) dt = \mathcal{T}/2$,

$$\Delta S = \frac{mA^2\mathcal{T}}{4} \left[\left(\frac{\pi}{\mathcal{T}} \right)^2 - \omega^2 \right].$$

For $\mathcal{T} < \pi/\omega$ this is positive and the stationary path is a minimum. For $\mathcal{T} > \pi/\omega$ — a trip lasting longer than *half a period* — it is **negative**: we have exhibited a nearby path with strictly smaller action. The stationary path is a saddle. Nothing has gone wrong; $\delta S = 0$ still holds, and the equation of motion is still right. "Least" was simply never the claim.

The instant $\mathcal{T} = \pi/\omega$ at which the minimum is lost is called a **conjugate point**: the time at which a second solution with the same endpoints first appears (here, the oscillator can start at the origin, swing out, and return exactly half a period later). ▀ The general theory — Jacobi's theorem, that the second variation is positive definite precisely up to the first conjugate point — is quoted, not proved. The clean geometric example is on a sphere: *both* arcs of a great circle joining two cities satisfy the geodesic equation, and the long way round is a perfectly good stationary point of arc length that is emphatically not the shortest path. In Chapter 3.7 those two arcs become the two images of a gravitationally lensed quasar, and the conjugate point becomes the caustic.

▼ Grind box — the second variation in general, and the one condition you can check by eye

The oscillator's ΔS was exact because its L was quadratic. In general it is only the leading correction, and getting it is Chapter 0.6's second-order Taylor expansion applied to $\Phi(\epsilon) = S[x + \epsilon\eta]$.

Differentiate (1.2.15) once more in ϵ and set $\epsilon = 0$:

$$\delta^2 S \equiv \Phi''(0) = \int_{t_1}^{t_2} \left[\frac{\partial^2 L}{\partial x^2} \eta^2 + 2 \frac{\partial^2 L}{\partial x \partial \dot{x}} \eta \dot{\eta} + \frac{\partial^2 L}{\partial \dot{x}^2} \dot{\eta}^2 \right] dt,$$

so that $S[x + \epsilon\eta] = S[x] + \epsilon \delta S + \frac{1}{2} \epsilon^2 \delta^2 S + o(\epsilon^2)$. This is exactly Chapter 0.6's Hessian, with the matrix replaced by an integral kernel. Tidy it using $2\eta\dot{\eta} = \frac{d}{dt}(\eta^2)$ and one integration by parts, whose boundary term dies as usual:

$$\delta^2 S = \int_{t_1}^{t_2} [P(t) \dot{\eta}^2 + Q(t) \eta^2] dt, \quad P = \frac{\partial^2 L}{\partial \dot{x}^2}, \quad Q = \frac{\partial^2 L}{\partial x^2} - \frac{d}{dt} \left(\frac{\partial^2 L}{\partial x \partial \dot{x}} \right).$$

The condition you can check by eye. A minimum requires $P(t) \geq 0$ everywhere along the path. The argument is the reason fast wiggles matter: take η to be a tiny bump of height A concentrated in a window of width w around a point where $P < 0$. Then $\int Q\eta^2 \sim A^2 w$ while $\int P\dot{\eta}^2 \sim PA^2/w$, so shrinking w makes the P term dominate with whatever sign P has. If $P < 0$ anywhere, $\delta^2 S$ can be made as negative as you like — no minimum. This is the **Legendre condition**, and for $L = T - V$ it reads $P = m > 0$, satisfied automatically.

It is also worth checking on the one action in §8.1's table you can already differentiate. Chapter 2.5's free relativistic particle has $L = -mc^2 \sqrt{1 - \dot{x}^2/c^2}$, giving $\partial L / \partial \dot{x} = \gamma m \dot{x}$ and $P = \partial^2 L / \partial \dot{x}^2 = m\gamma^3 > 0$. So S is minimised — and since $S = -mc^2 \int d\tau$ is *minus* the proper time, minimising S means **maximising the proper time** along the worldline. The minus sign in front of mc^2 is there precisely to arrange that, and it is why the travelling twin, who takes a non-extremal path, comes back younger.

▮ Legendre's condition is necessary, not sufficient. The sufficient statement — Jacobi's theorem, that $\delta^2 S > 0$ for all nonzero admissible η precisely when $P > 0$ and the interval contains no conjugate point — is quoted here and not proved; it is a boundary-value problem for the equation $\frac{d}{dt}(P\dot{u}) - Qu = 0$, whose zeros are the conjugate points. For the oscillator, $P = m$, $Q = -m\omega^2$, that equation is $\ddot{u} = -\omega^2 u$, its solution starting at $u(0) = 0$ is $\sin \omega t$, and its next zero is at $t = \pi/\omega$ — exactly the half-period the callout found by exhibiting a variation. The general theory reproduces the special case, which is the least one can ask of it.

6 · Constraints become trivial — the payoff

So far the action principle has cost effort and returned Newton's laws, which we already had. This section is where it starts paying.

6.1 · Generalized coordinates

A **generalized coordinate system** is any set of n numbers $q^1, \dots, q^n$ that fixes the configuration of the system, where n is the number of independent ways the system can move. The q 's need not be lengths. They can be angles, areas, arc lengths along a wire, the amplitude of a normal mode, or anything else, and they need not have the same dimensions as each other.

Why this is allowed is the whole point of §3: the derivation of (1.2.23) never assumed anything about what the q^i meant. It assumed only that they could be varied independently. So the recipe is:

1. Choose coordinates that *automatically satisfy the constraints* — one coordinate per genuine degree of freedom, so that every value of the q^i describes an allowed configuration and every allowed configuration corresponds to some q^i .

2. Write T and V in terms of them.
3. Turn the crank (1.2.23).

Constraint forces never appear, because in step 1 the constraint stopped being a force and became a coordinate.

6.2 · The pendulum, in three lines

A bob of mass m on a rigid massless rod of length ℓ , swinging in a vertical plane about a fixed pivot. Two Cartesian coordinates, one rigid constraint, so $n = 1$; take θ , the angle from the downward vertical.

The bob moves on a circle of radius ℓ , so its speed is $\ell\dot{\theta}$ and its height relative to the pivot is $-\ell \cos \theta$. Hence

$$T = \frac{1}{2}m\ell^2\dot{\theta}^2, \quad V = -mg\ell \cos \theta, \quad L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta. \quad (1.2.36)$$

Two derivatives:

$$\frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta}, \quad \frac{\partial L}{\partial \theta} = -mg\ell \sin \theta. \quad (1.2.37)$$

Feed them to (1.2.22):

$$\frac{d}{dt}(m\ell^2\dot{\theta}) + mg\ell \sin \theta = 0 \quad \Longleftrightarrow \quad \ddot{\theta} = -\frac{g}{\ell} \sin \theta. \quad (1.2.38)$$

That is the whole derivation. No forces were resolved, no free-body diagram was drawn, and — the point — **the tension in the rod never appeared**. Chapter 1.1 got the same equation by resolving forces along and perpendicular to the rod, which required introducing the unknown tension $\mathcal{F}$, writing two component equations, and eliminating $\mathcal{F}$ between them. Worked Example 1 does that side by side so you can count the lines.

Why is the tension absent here? Because the rod's tension is always perpendicular to the bob's motion, so it **does no work**. Energies are what L is built from, and a force that does no work contributes no energy, so it cannot appear in L . Meanwhile the geometric job the tension was doing — keeping the bob at distance ℓ — has been absorbed into the choice of θ as the coordinate, since every value of θ automatically has the bob at radius ℓ . The constraint has been converted from a force into a coordinate, and that trade is the central practical advantage of the Lagrangian formulation.

6.3 · A harder one, to show the method scales

Now the double pendulum: a rod of length ℓ_1 from a fixed pivot to mass m_1 , and a second rod of length ℓ_2 from m_1 to m_2 . Four Cartesian coordinates, two rigid constraints, so $n = 2$; take θ_1 and θ_2 , each measured from the downward vertical.

Positions, with y measured downward from the pivot:

$$\begin{aligned}x_1 &= \ell_1 \sin \theta_1, & y_1 &= \ell_1 \cos \theta_1, \\x_2 &= \ell_1 \sin \theta_1 + \ell_2 \sin \theta_2, & y_2 &= \ell_1 \cos \theta_1 + \ell_2 \cos \theta_2.\end{aligned}\tag{1.2.39}$$

Differentiate and square. For the second mass, $\dot{x}_2^2 + \dot{y}_2^2 = \ell_1^2 \dot{\theta}_1^2 + \ell_2^2 \dot{\theta}_2^2 + 2\ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 (\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2)$, and the bracket collapses by the angle-subtraction identity. Heights are $-y$, so

$$\begin{aligned}L &= \frac{1}{2}(m_1 + m_2)\ell_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 \ell_2^2 \dot{\theta}_2^2 + m_2 \ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \\&\quad + (m_1 + m_2)g\ell_1 \cos \theta_1 + m_2 g \ell_2 \cos \theta_2.\end{aligned}\tag{1.2.40}$$

Two lines, and the dynamics is completely specified: apply (1.2.23) twice and you have the equations of motion, coupling and all. The Newtonian route requires four unknown constraint-force components (two rods, each exerting a force with two components on each of its ends), four vector equations, and the elimination of all four unknowns. The Lagrangian route needed no unknowns at all, because the coordinates (θ_1, θ_2) already encode both rigid rods.

This is the scaling argument. Adding a third link adds one coordinate, one term to T , one term to V , and one Euler–Lagrange equation. In the force picture it adds two more unknown constraint forces to be eliminated. The Lagrangian method grows linearly in the number of degrees of freedom; the force method grows with the number of constraints, which is exactly the quantity you were hoping to make go away.

▼ Grind box — the double pendulum's kinetic energy, every step

Differentiate (1.2.39):

$$\begin{aligned}\dot{x}_1 &= \ell_1 \dot{\theta}_1 \cos \theta_1, & \dot{y}_1 &= -\ell_1 \dot{\theta}_1 \sin \theta_1, \\ \dot{x}_2 &= \ell_1 \dot{\theta}_1 \cos \theta_1 + \ell_2 \dot{\theta}_2 \cos \theta_2, & \dot{y}_2 &= -\ell_1 \dot{\theta}_1 \sin \theta_1 - \ell_2 \dot{\theta}_2 \sin \theta_2.\end{aligned}$$

The first mass is easy: $\dot{x}_1^2 + \dot{y}_1^2 = \ell_1^2 \dot{\theta}_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) = \ell_1^2 \dot{\theta}_1^2$. For the second, square both and add:

$$\begin{aligned}\dot{x}_2^2 &= \ell_1^2 \dot{\theta}_1^2 \cos^2 \theta_1 + \ell_2^2 \dot{\theta}_2^2 \cos^2 \theta_2 + 2\ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 \cos \theta_1 \cos \theta_2, \\ \dot{y}_2^2 &= \ell_1^2 \dot{\theta}_1^2 \sin^2 \theta_1 + \ell_2^2 \dot{\theta}_2^2 \sin^2 \theta_2 + 2\ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 \sin \theta_1 \sin \theta_2.\end{aligned}$$

The Pythagorean identity collapses the first two columns, and the cross terms combine through $\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 = \cos(\theta_1 - \theta_2)$:

$$\dot{x}_2^2 + \dot{y}_2^2 = \ell_1^2 \dot{\theta}_1^2 + \ell_2^2 \dot{\theta}_2^2 + 2\ell_1 \ell_2 \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2).$$

Then $T = \frac{1}{2}m_1 \ell_1^2 \dot{\theta}_1^2 + \frac{1}{2}m_2 (\dot{x}_2^2 + \dot{y}_2^2)$, which on collecting the $\dot{\theta}_1^2$ terms gives (1.2.40). With y measured downward, the height of mass j above the pivot is $-y_j$, so $V = -m_1 g y_1 - m_2 g y_2$, giving the two potential terms with the signs shown.

A structural remark you will need in Chapter 1.3. The kinetic energy came out as a *quadratic form* in the generalized velocities, $T = \frac{1}{2} \sum_{ij} M_{ij}(\theta) \dot{\theta}^i \dot{\theta}^j$ with

$$M(\theta) = \begin{pmatrix} (m_1 + m_2)\ell_1^2 & m_2\ell_1\ell_2 \cos(\theta_1 - \theta_2) \\ m_2\ell_1\ell_2 \cos(\theta_1 - \theta_2) & m_2\ell_2^2 \end{pmatrix},$$

symmetric and positive definite — the "mass matrix". That is not an accident of this example.

Whenever the coordinate change from Cartesians is time-independent, T is *always* a positive-definite quadratic form in $\dot{q}$, because it is $\frac{1}{2} \sum m_a |\dot{\mathbf{x}}_a|^2$ pushed through a linear map (Chapter 0.5). Positive-definiteness is what makes $p = \partial L / \partial \dot{q}$ invertible for $\dot{q}$, which is precisely the condition Chapter 1.3 needs for the Legendre transform to exist. And a metric-like symmetric positive-definite matrix that depends on position is exactly the object Chapter 3.3 calls a **metric** — the kinetic energy is a metric on configuration space, and free motion is a geodesic in it.

6.4 · When you *want* the constraint force: Lagrange multipliers

Sometimes the constraint force is the answer you need — you are designing the rod and want to know whether it will snap. Eliminating it was convenient; now we want it back. Chapter 0.6 already built the tool.

There, the problem was to make $f(\mathbf{x})$ stationary subject to $g(\mathbf{x}) = 0$, and the resolution was to make the *unconstrained* function $f - \lambda(g - c)$ stationary in both $\mathbf{x}$ and λ , trading a restriction on the search space for one extra equation and one extra unknown. Do exactly the same thing here, with the search space now a space of paths.

Suppose we use redundant coordinates $q^1, \dots, q^N$ (N larger than the number of degrees of freedom) subject to a constraint $g(q, t) = 0$ that must hold at every instant. Consider the modified action

$$\tilde{S}[q, \lambda] = \int_{t_1}^{t_2} [L(q, \dot{q}, t) + \lambda(t) g(q, t)] dt, \quad (1.2.41)$$

where $\lambda(t)$ — note: a *function* of time, one multiplier per instant, because the constraint must hold at every instant — is now one more variable to be varied over. Varying with respect to λ is easy: λ appears without its derivative, so (1.2.22) with $\partial \tilde{L} / \partial \dot{\lambda} = 0$ gives $\partial \tilde{L} / \partial \lambda = g(q, t) = 0$. The constraint has been recovered as an equation of motion. Varying with respect to q^i gives

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}^i} \right) - \frac{\partial L}{\partial q^i} = \lambda \frac{\partial g}{\partial q^i}. \quad (1.2.42)$$

Compare with (1.2.23). The left-hand side is the free Euler–Lagrange expression, which in Cartesian coordinates is $\dot{p}_i - F_i$; so the right-hand side is *an extra force*, and $\lambda \partial g / \partial q^i$ is precisely the force that enforces the constraint. Two features are worth naming:

- The constraint force points along ∇g — perpendicular to the constraint surface, exactly as Chapter 0.6 showed the gradient is normal to a level set. That is the mathematical content of "the constraint force does no work": motion allowed by the constraint is tangent to the surface, and a force normal to it has zero dot product with the motion.
- The multiplier **is** the magnitude of that force, up to the normalisation of g — which matches Chapter 0.6's finding that $\lambda = df^*/dc$ is the sensitivity of the optimum to relaxing the constraint. Relax a rod's length and the work you must do is set by its tension. Same λ , same meaning.

Worked Example 1 runs this on the pendulum and recovers the tension explicitly. The same trade — restrict the space, or enlarge it and add a multiplier — reappears in Chapter 3.6 (where the metric is varied subject to nothing, but the analogous structure produces constraint equations) and in Chapter 6.3, where the "constraint" being imposed by a multiplier is a gauge condition.

7 · Form invariance — the property that survives to general relativity

Here is the structural property that makes the Lagrangian formulation not merely convenient but *necessary* for everything after Chapter 2.

Claim. Let $q^i = q^i(Q^1, \dots, Q^n, t)$ be any invertible, smooth change of coordinates (possibly time-dependent), and define the Lagrangian in the new coordinates by substitution,

$$\tilde{L}(Q, \dot{Q}, t) \equiv L(q(Q, t), \dot{q}(Q, \dot{Q}, t), t). \quad (1.2.43)$$

Then $Q^a(t)$ satisfies the Euler–Lagrange equations of $\tilde{L}$ if and only if $q^i(t)$ satisfies the Euler–Lagrange equations of L .

The one-line reason. S is a number attached to a path. Relabelling which numbers you use to name the points of the path cannot change that number — you fed in the same physical path and got the same integral. So the *function* $\Phi(\epsilon) = S[\text{path deformed by } \epsilon]$ is identical in both descriptions, and so is the statement $\Phi'(0) = 0$. Stationarity is a property of the path, not of the names.

That argument is correct and it is the reason. But it is worth also seeing the derivative-level version, because the two lemmas it uses recur constantly.

Lemma 1 (cancellation of the dots). Differentiating $q^i(Q(t), t)$ along the path,

$$\dot{q}^i = \sum_a \frac{\partial q^i}{\partial Q^a} \dot{Q}^a + \frac{\partial q^i}{\partial t}. \quad (1.2.44)$$

The right-hand side is *linear* in $\dot{Q}^a$ with coefficients depending only on Q and t . Reading off the coefficient,

$$\frac{\partial \dot{q}^i}{\partial \dot{Q}^a} = \frac{\partial q^i}{\partial Q^a}. \quad (1.2.45)$$

Lemma 2 (partial and total derivatives commute). Differentiate (1.2.44) with respect to Q^a at fixed $\dot{Q}$; separately take the total time derivative of $\partial q^i / \partial Q^a$ using the chain rule. Both give the same thing:

$$\frac{\partial \dot{q}^i}{\partial Q^a} = \sum_b \frac{\partial^2 q^i}{\partial Q^a \partial Q^b} \dot{Q}^b + \frac{\partial^2 q^i}{\partial Q^a \partial t} = \frac{d}{dt} \left(\frac{\partial q^i}{\partial Q^a} \right), \quad (1.2.46)$$

where the middle step used the equality of mixed partials (Chapter 0.6). Now the computation. From (1.2.43) and the chain rule, using Lemma 1,

$$\frac{\partial \tilde{L}}{\partial \dot{Q}^a} = \sum_i \frac{\partial L}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial \dot{Q}^a} = \sum_i \frac{\partial L}{\partial \dot{q}^i} \frac{\partial q^i}{\partial Q^a}, \quad (1.2.47)$$

and, since $\tilde{L}$ depends on Q^a both directly and through $\dot{q}$,

$$\frac{\partial \tilde{L}}{\partial Q^a} = \sum_i \left[\frac{\partial L}{\partial q^i} \frac{\partial q^i}{\partial Q^a} + \frac{\partial L}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial Q^a} \right]. \quad (1.2.48)$$

Take d/dt of (1.2.47) with the product rule and subtract (1.2.48):

$$\begin{aligned} \frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}^a} - \frac{\partial \tilde{L}}{\partial Q^a} &= \sum_i \left\{ \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} \right] \frac{\partial q^i}{\partial Q^a} + \frac{\partial L}{\partial \dot{q}^i} \frac{d}{dt} \left(\frac{\partial q^i}{\partial Q^a} \right) \right. \\ &\quad \left. - \frac{\partial L}{\partial q^i} \frac{\partial q^i}{\partial Q^a} - \frac{\partial L}{\partial \dot{q}^i} \frac{\partial \dot{q}^i}{\partial Q^a} \right\}. \end{aligned} \quad (1.2.49)$$

By Lemma 2 the second and fourth terms are equal and cancel. What is left factorises:

$$\boxed{\frac{d}{dt} \frac{\partial \tilde{L}}{\partial \dot{Q}^a} - \frac{\partial \tilde{L}}{\partial Q^a} = \sum_i \left[\frac{d}{dt} \frac{\partial L}{\partial \dot{q}^i} - \frac{\partial L}{\partial q^i} \right] \frac{\partial q^i}{\partial Q^a}.} \quad (1.2.50)$$

The Euler–Lagrange expressions transform into each other by multiplication by the Jacobian matrix $\partial q^i / \partial Q^a$, which is invertible because the coordinate change is. So one set vanishes exactly when the other does. ■

7.1 • Why this matters more than it looks

Now the contrast. $F = ma$ is **not form invariant**. Write Newton's law in plane polar coordinates and it does not read $m\ddot{r} = F_r$. The acceleration of a particle in polar coordinates has extra pieces:

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2) \hat{\mathbf{r}} + (r\ddot{\theta} + 2\dot{r}\dot{\theta}) \hat{\boldsymbol{\theta}}. \quad (1.2.51)$$

The $-r\dot{\theta}^2$ is the centrifugal term and the $2\dot{r}\dot{\theta}$ is the Coriolis term, and they arise because the basis vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}$ themselves rotate as the particle moves, so differentiating a vector twice requires differentiating the basis too. Getting (1.2.51) honestly takes half a page of unit-vector derivatives.

Now get it from the Lagrangian. In polar coordinates $T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2)$ — just Pythagoras on the radial and tangential velocity components — so with a potential $V(r, \theta)$,

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - V(r, \theta). \quad (1.2.52)$$

Turn the crank. For $q = r$: $\partial L / \partial \dot{r} = m\dot{r}$ and $\partial L / \partial r = mr\dot{\theta}^2 - \partial V / \partial r$, so

$$m\ddot{r} - mr\dot{\theta}^2 = -\frac{\partial V}{\partial r}. \quad (1.2.53)$$

For $q = \theta$: $\partial L / \partial \dot{\theta} = mr^2\dot{\theta}$ and $\partial L / \partial \theta = -\partial V / \partial \theta$, so

$$\frac{d}{dt}(mr^2\dot{\theta}) = -\frac{\partial V}{\partial \theta} \quad \Longleftrightarrow \quad m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -\frac{1}{r}\frac{\partial V}{\partial \theta}. \quad (1.2.54)$$

Both centrifugal and Coriolis terms appeared automatically, from differentiating a scalar. No unit vectors were differentiated; no fictitious forces had to be remembered. And (1.2.54) hands you a bonus: if V does not depend on θ , the left side says $mr^2\dot{\theta}$ is constant. Angular momentum conservation, for free, because θ was missing from L .

WHY THIS DECIDES THE SHAPE OF THE REST OF THE BOOK

Newton's second law is a relation between *vector components*, and vector components transform in a complicated way when the basis varies from point to point. The law therefore has a privileged coordinate system — Cartesian — in which it is simple, and grows correction terms everywhere else.

General relativity has **no** global Cartesian coordinates. Not "it's awkward to find them"; there provably are none, because spacetime is curved, and Chapter 3.4 will make that a precise statement about a tensor that cannot be made to vanish. A formulation of physics that is simple only in a coordinate system that does not exist is not a formulation of physics. It is a special case.

The action is a *scalar*: a single number per path, with no components to transform. (1.2.50) says its stationarity condition therefore keeps the same shape in every coordinate system. That is why Chapters 2.6, 3.6, 5.2 and 6.4 all consist of the same sentence — "write down the most general scalar action invariant under the symmetry you believe in, and vary it" — and why they never once resolve a force.

FAMILIAR GROUND

You already run a variational principle. When you fit a model by maximum likelihood you do not write down an estimator. You write down a *scalar functional of the parameters*, the log-likelihood $\ell(\theta)$, declare that the answer is where it is stationary, and let the stationarity condition

$$\frac{\partial \ell}{\partial \theta} = 0$$

— the score equation — *produce* the estimator. For a normal mean it produces the sample mean; for a Cox model it produces something you would never have guessed. Chapter 0.6 used the second derivative of that same surface: the Hessian at the peak is the observed information, and its inverse is the covariance matrix of the estimate. Here we are using the first derivative instead, and the parallel is exact: **specify a scalar to be made stationary, and the equations you actually wanted are its stationarity condition.** The Euler–Lagrange equation is the score equation of physics. The Hessian's role — telling you whether the stationary point is a maximum, a minimum, or neither — is played here by the second variation, which is exactly what the harmonic-oscillator warning callout computed.

The correspondence goes further than an analogy, into literally the same mathematics, the moment your unknown is a function rather than a parameter vector. Fit a smooth dose–response curve by penalised least squares: over *all* functions f , minimise

$$J[f] = \sum_j (y_j - f(x_j))^2 + \lambda \int_a^b f''(x)^2 dx.$$

That is a functional on an infinite-dimensional space, and finding its minimiser is the calculus of variations, not statistics. Vary $f \rightarrow f + \epsilon \eta$; the penalty term contributes $2\lambda \int f'' \eta'' dx$, and integrating by parts *twice* — the same move as §3.3, performed once more — turns $\int f'' \eta''$ into $\int f'''' \eta$ plus boundary terms. Between data points, where the first sum contributes nothing, the Euler–Lagrange equation is therefore the one derived in §3.5's grind box with $F = \lambda f''^2$, namely

$$f''''(x) = 0,$$

whose solutions are exactly the cubic polynomials. *That* is why the answer is a cubic spline — not because someone chose cubics, but because they are the kernel of the Euler–Lagrange operator of the roughness penalty. And the boundary terms that by-parts leaves behind, having no fixed-endpoint condition to kill them, must vanish on their own: $f'' = f''' = 0$ at the outer knots, which forces the fitted curve to be *linear* beyond them. If you have ever wondered why a restricted cubic spline is constrained to be linear in the tails, that is the answer, and it is a boundary term in a calculus-of-variations problem. Same manoeuvre, same integration by parts, same vanishing boundary term as §3.3 — only the interpretation differs.

8 · One equation to rule them all — a preview

Everything so far has a path $x(t)$: one independent variable, t , and finitely many dependent ones. Physics after Chapter 5 has *fields*: a quantity ϕ defined at every point of spacetime, so that the dependent variable is indexed by four independent ones, $x^\mu = (t, x, y, z)$.

The generalisation writes itself. Replace the Lagrangian by a Lagrangian *density* $\mathcal{L}(\phi, \partial_\mu \phi)$ and the time integral by an integral over a region Ω of spacetime:

$$S[\phi] = \int_{\Omega} \mathcal{L}(\phi, \partial_\mu \phi) d^4x. \quad (1.2.55)$$

Now repeat §3 *verbatim*. Vary $\phi \rightarrow \phi + \epsilon\eta$ with η vanishing on the boundary $\partial\Omega$ instead of at two endpoints. Differentiate under the integral sign. Linearize the integrand by the chain rule:

$$\delta S = \int_{\Omega} \left[\frac{\partial \mathcal{L}}{\partial \phi} \eta + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \partial_\mu \eta \right] d^4x. \quad (1.2.56)$$

Integrate by parts to move the derivative off $\partial_\mu \eta$ — in four dimensions "by parts" means the divergence theorem of Chapter 0.7, which converts $\int_{\Omega} \partial_\mu (\dots)^\mu d^4x$ into a surface integral over $\partial\Omega$. That surface term dies because η vanishes there — same reason, same sentence. Apply the fundamental lemma, whose proof needs only the obvious four-dimensional bump function. Out comes

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0.} \quad (1.2.57)$$

Put (1.2.57) beside (1.2.22). The dictionary is $t \rightarrow x^\mu$, $dt \rightarrow d^4x$, $\frac{d}{dt} \rightarrow \partial_\mu$, $\dot{x} \rightarrow \partial_\mu \phi$, and nothing else. **The derivation you did in §3 is the derivation that produces every field equation in physics.** Chapter 5.2 develops this properly, including what happens when ϕ carries indices and what Lorentz invariance demands of $\mathcal{L}$; there is no point duplicating it here. The point is that there is nothing new to learn — only the same four moves in more dimensions.

8.1 · The actions this book will write down

► Everything in this table is **quoted forward**. Not one line is derived here; each is constructed, from symmetry principles, in the chapter named. It is on this page because seeing the list once, early, tells you what kind of book this is: after Chapter 1.2, physics consists of choosing a scalar and varying it.

SYSTEM	ACTION	CHAPTER
Free particle	$S = \int \frac{1}{2} m \dot{\mathbf{x}}^2 dt$	§4
Relativistic particle	$S = -mc^2 \int d\tau$	2.5
Electromagnetic field	$S = -\frac{1}{4} \int F_{\mu\nu} F^{\mu\nu} d^4x$	2.6
Gravity (Einstein–Hilbert)	$S = \frac{1}{2\kappa} \int R \sqrt{-g} d^4x$	3.6
Scalar field (Klein–Gordon)	$S = \int (\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2) d^4x$	5.3
Spin- $1/2$ field (Dirac)	$S = \int \bar{\psi} (i\gamma^\mu \partial_\mu - m) \psi d^4x$	5.4
Non-abelian gauge field (Yang–Mills)	$S = -\frac{1}{2} \int \text{tr} (F_{\mu\nu} F^{\mu\nu}) d^4x$	6.4
Relativistic string (Polyakov)	$S = -\frac{T}{2} \int \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X_\mu d^2\sigma$	7.3

Eight lines. Between them they contain classical mechanics, special relativity, electromagnetism, gravitation, quantum field theory, the strong and electroweak interactions, and string theory. Every one is fed to (1.2.22) or (1.2.57), and the output is the equations of motion of that theory. The rest of this book is, to a first approximation, the story of how each of those eight lines gets *chosen* — which is always the same story: pick a symmetry, and write the most general scalar invariant under it.

9 · Worked examples

WORKED EXAMPLE 1 — THE PENDULUM, THREE WAYS, WITH THE TENSION RECOVERED

(a) The Newtonian route, as in Chapter 1.1. The bob has two forces on it: gravity $m\mathbf{g}$ downward, and the rod tension $\mathcal{F}$ directed along the rod toward the pivot. Resolve along the two natural directions. Tangentially (in the direction of increasing θ), gravity contributes $-mg \sin \theta$ and the tension contributes nothing, while the tangential acceleration is $\ell\ddot{\theta}$:

$$m\ell\ddot{\theta} = -mg \sin \theta.$$

Radially (toward the pivot), the acceleration is the centripetal $\ell\dot{\theta}^2$, and the inward forces are $\mathcal{F}$ minus the inward component of gravity $mg \cos \theta$:

$$m\ell\dot{\theta}^2 = \mathcal{F} - mg \cos \theta.$$

Two equations, two unknowns ($\theta, \mathcal{F}$). The first is what we wanted; the second exists only to determine $\mathcal{F}$, which we then discard. To write either of them you had to know in advance that the acceleration in polar coordinates has a centripetal piece — that is, you had to have (1.2.51) already.

(b) The Lagrangian route. One coordinate, θ . Kinetic energy $\frac{1}{2}m\ell^2\dot{\theta}^2$, potential $-mg\ell \cos \theta$,

$$\begin{aligned} L &= \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell \cos \theta \\ \implies \frac{d}{dt}(m\ell^2\dot{\theta}) + mg\ell \sin \theta &= 0 \implies \ddot{\theta} = -\frac{g}{\ell} \sin \theta. \end{aligned}$$

One unknown, one equation, no tension, no unit-vector derivatives, no (1.2.51). The count is: (a) needed two vector resolutions and one elimination; (b) needed two partial derivatives.

(c) Recovering the tension with a multiplier. Now deliberately *do not* impose the constraint. Use polar coordinates (r, θ) in the plane, so that

$$L = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + mgr \cos \theta,$$

with the constraint $g(r) = r - \ell = 0$. Following (1.2.41), vary $\tilde{L} = L + \lambda(t)(r - \ell)$. The λ equation returns $r = \ell$. The θ equation, with r then set to ℓ , returns the pendulum equation again. The r equation is the new information:

$$\frac{d}{dt}(m\dot{r}) - (mr\dot{\theta}^2 + mg \cos \theta + \lambda) = 0.$$

On the constraint surface $r \equiv \ell$, so $\dot{r} = \ddot{r} = 0$, and

$$\lambda = -m(\ell\dot{\theta}^2 + g \cos \theta).$$

The generalized constraint force is $\lambda \partial g / \partial r = \lambda$, and it is *negative* — directed toward decreasing r , i.e. toward the pivot, which is exactly the direction a rod pulls. Its magnitude is

$$\mathcal{F} = |\lambda| = m(\ell\dot{\theta}^2 + g \cos \theta),$$

identical to what route (a) produced. The multiplier is the tension. Note the two contributions, both physically sensible: $mg \cos \theta$ holds up the component of weight along the rod, and $m\ell\dot{\theta}^2$ supplies the centripetal force. At the bottom of a swing released from horizontal, energy conservation gives $\ell\dot{\theta}^2 = 2g$ so $\mathcal{F} = 3mg$ — the standard result, obtained here without ever drawing a free-body diagram.

Small oscillations, since we are here. For $|\theta| \ll 1$, $\sin \theta \approx \theta$ (Chapter 0.3) and (1.2.38) becomes $\ddot{\theta} = -(g/\ell)\theta$: the harmonic oscillator of Chapter 0.8, with $\omega = \sqrt{g/\ell}$ and period $2\pi\sqrt{\ell/g}$ — the formula Chapter 0.1 differentiated logarithmically in its first worked example, now derived rather than quoted.

WORKED EXAMPLE 2 — GEODESICS: THE STRAIGHT LINE, AND THE GREAT CIRCLE

Strip away the physics. Take the arc-length functional (1.2.1) and ask which curve makes it stationary. This is the purest possible variational problem — no masses, no forces, no time — and it is the one that becomes gravity.

(a) In the plane. Minimise $\mathcal{A}[y] = \int_a^b \sqrt{1 + y'^2} \, dx$ between fixed endpoints. Here x plays the role of t and y the role of the coordinate, so the Euler–Lagrange equation (1.2.22) applies with $F(y, y') = \sqrt{1 + y'^2}$. Since F does not contain y at all,

$$\frac{\partial F}{\partial y} = 0 \quad \implies \quad \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0 \quad \implies \quad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}} = c$$

for some constant c . Do not stop there — the equation still has to be solved for y' . Square both sides and clear the denominator:

$$y'^2 = c^2(1 + y'^2) \quad \implies \quad y'^2(1 - c^2) = c^2 \quad \implies \quad y' = \frac{c}{\sqrt{1 - c^2}} \equiv m,$$

a constant. (The division is legitimate: $|c| = |y'| / \sqrt{1 + y'^2} < 1$ for every real y' , so $1 - c^2 > 0$ always.)

Integrating, $y = mx + b$. **The straight line, from a variational principle** — and note that we never assumed the answer was a line, or that it was smooth beyond C^1 , or anything else. We asked for stationarity and got straightness.

(b) On a sphere. Now do the identical calculation with a different notion of distance. On the unit sphere with polar angle θ (from the north pole) and azimuth ϕ , an infinitesimal displacement has length

$$ds^2 = d\theta^2 + \sin^2 \theta \, d\phi^2,$$

because moving in θ traverses a great circle of radius 1 while moving in ϕ traverses a circle of latitude of radius $\sin \theta$. Parametrising the curve as $\phi(\theta)$,

$$\mathcal{A}[\phi] = \int_{\theta_1}^{\theta_2} \sqrt{1 + \sin^2 \theta \, \phi'^2} \, d\theta, \quad F(\theta, \phi') = \sqrt{1 + \sin^2 \theta \, \phi'^2}.$$

Again F contains no ϕ , so again $\partial F / \partial \phi'$ is constant:

$$\frac{\sin^2 \theta \, \phi'}{\sqrt{1 + \sin^2 \theta \, \phi'^2}} = k.$$

Take the case $k = 0$ first. Since $\sin \theta \neq 0$ away from the poles, $k = 0$ forces $\phi' = 0$, so $\phi = \text{constant}$: the curve is a **meridian**, which is a great circle.

That single case is enough, because of a symmetry we can check directly. The metric $d\theta^2 + \sin^2 \theta \, d\phi^2$ is the ordinary Euclidean length restricted to the sphere $|\mathbf{r}| = 1$, and rotations of $\mathbb{R}^3$ preserve Euclidean length and map the sphere to itself — so a rotation carries curves to curves of the same length, and therefore carries stationary curves to stationary curves. Given any two points on the sphere, rotate so that both lie on a common meridian. The stationary curve joining them is then a meridian arc; rotating back, the stationary curve joining the original two points is the image of a meridian, which is a **great circle**. ■ (The general $k \neq 0$ case can also be integrated directly, and gives the same answer; see the grind box.)

What just happened. The *same functional*, $\int ds$, gave a straight line in one geometry and a great circle in another. The only input that changed was the formula for ds — the **metric**. The variational principle did not change at all.

Chapter 3.3 will write $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$ for a metric on curved *spacetime*, feed $\int ds$ into exactly this machinery, and call the resulting Euler–Lagrange equation the **geodesic equation**. A planet orbiting the sun will turn out to be doing what the great circle does — travelling as straight as the geometry permits. Gravity will not be a force in that description; it will be the metric. Nothing about the method in this worked example will change. Only ds .

▼ Grind box — integrating the sphere's geodesic equation in the general case

Solve $\sin^2 \theta \phi' / \sqrt{1 + \sin^2 \theta \phi'^2} = k$ for ϕ' . Square and cross-multiply:

$$\sin^4 \theta \phi'^2 = k^2 (1 + \sin^2 \theta \phi'^2) \implies \phi'^2 \sin^2 \theta (\sin^2 \theta - k^2) = k^2 \implies \phi' = \frac{k}{\sin \theta \sqrt{\sin^2 \theta - k^2}}.$$

Substitute $u = \cot \theta$, so $du = -\csc^2 \theta d\theta$ and $\sin^2 \theta - k^2 = \sin^2 \theta (1 - k^2(1 + u^2))$. The integral collapses:

$$\phi - \phi_0 = \int \frac{k d\theta}{\sin^2 \theta \sqrt{1 - k^2 - k^2 u^2}} = - \int \frac{k du}{\sqrt{1 - k^2 - k^2 u^2}} = - \int \frac{du}{\sqrt{A^2 - u^2}}, \quad A^2 \equiv \frac{1 - k^2}{k^2}.$$

Hence $\phi - \phi_0 = -\arcsin(u/A)$, i.e. $\cot \theta = -A \sin(\phi - \phi_0)$. Multiply through by $\sin \theta$ and expand the sine:

$$\cos \theta = -A \sin \theta (\sin \phi \cos \phi_0 - \cos \phi \sin \phi_0).$$

Now recognise the Cartesian coordinates of a point on the unit sphere: $X = \sin \theta \cos \phi$, $Y = \sin \theta \sin \phi$, $Z = \cos \theta$. The equation becomes

$$(A \sin \phi_0)X - (A \cos \phi_0)Y - Z = 0,$$

which is a **plane through the origin**. A plane through the centre of a sphere cuts it in a great circle. So every solution, for every k , is a great circle — the $k = 0$ case being the limit in which the plane contains the polar axis. (Verified numerically to 7×10^{-16} for $A = 0.8$, $\phi_0 = 0.4$.)

Notice that $k = \partial F / \partial \phi'$ is conserved because ϕ is absent from F — the same structure that gave angular momentum conservation in (1.2.54). On a sphere it is the statement that a geodesic's bearing satisfies $\sin \theta \cos(\text{bearing}) = \text{constant}$, which navigators call *Clairaut's relation* and which is the reason a great-circle route from London to Tokyo goes over the Arctic. It is Noether's theorem (Chapter 1.4), arriving three chapters early disguised as cartography.

10 · Your turn

PROBLEM 1 — THE BRACHISTOCHRONE

A bead slides without friction down a wire from the origin to a lower point, starting from rest, under gravity g . Let y be measured *downward*. Set up the travel-time functional, reduce it with the fact that the integrand has no explicit x -dependence, and integrate the resulting equation to find the curve. (This is the problem Johann Bernoulli posed to Europe in 1696; Newton is said to have solved it overnight and returned it anonymously, whereupon Bernoulli reportedly recognised the author "as the lion by his claw".)

– Solution

Set-up. Energy conservation from rest gives $\frac{1}{2}mv^2 = mgy$, so $v = \sqrt{2gy}$ — independent of the shape of the wire, which is what makes the problem tractable. The time to traverse an element of arc is ds/v , so

$$\mathcal{T}[y] = \int_0^{x_1} \frac{\sqrt{1+y'^2}}{\sqrt{2gy}} dx \equiv \int_0^{x_1} F(y, y') dx.$$

Note this is (1.2.2) with refractive index $n \propto 1/\sqrt{y}$: the brachistochrone is literally a ray of light in a medium whose index falls off as the inverse square root of depth. Fermat and Bernoulli are the same problem.

Reduce. F has no explicit x , so by Problem 2's Beltrami identity the quantity $y'F_{y'} - F$ is constant along the solution. Compute it:

$$F_{y'} = \frac{y'}{\sqrt{2gy}\sqrt{1+y'^2}}, \quad y'F_{y'} - F = \frac{y'^2 - (1+y'^2)}{\sqrt{2gy}\sqrt{1+y'^2}} = \frac{-1}{\sqrt{2gy}\sqrt{1+y'^2}}.$$

Setting this equal to $-1/\sqrt{2gC}$ for a constant C and squaring gives the beautifully compact first integral

$$y(1+y'^2) = C.$$

Integrate. Solve for y' and separate:

$$y' = \sqrt{\frac{C-y}{y}} \quad \Rightarrow \quad dx = \sqrt{\frac{y}{C-y}} dy.$$

The substitution that rationalises this is $y = \frac{C}{2}(1 - \cos \vartheta) = C \sin^2(\vartheta/2)$, whence $C - y = C \cos^2(\vartheta/2)$, the square root becomes $\tan(\vartheta/2)$, and $dy = C \sin(\vartheta/2) \cos(\vartheta/2) d\vartheta$. Multiply:

$$dx = \tan \frac{\vartheta}{2} \cdot C \sin \frac{\vartheta}{2} \cos \frac{\vartheta}{2} d\vartheta = C \sin^2 \frac{\vartheta}{2} d\vartheta = \frac{C}{2}(1 - \cos \vartheta) d\vartheta.$$

Integrating with $x = 0$ at $\vartheta = 0$ and writing $a = C/2$:

$$x = a(\vartheta - \sin \vartheta), \quad y = a(1 - \cos \vartheta).$$

These are the parametric equations of a **cycloid** — the curve traced by a point on the rim of a wheel of radius a rolling along the underside of the horizontal axis. The constant a is fixed by requiring the curve to pass through the target point.

Check. For the half-cycloid with $a = \frac{1}{2}$, running from the origin to $(\pi/2, 1)$ and $g = 9.8$: numerical quadrature gives a descent time of 0.7096 s along the cycloid versus 0.8406 s along the straight line between the same two points — 16% faster despite being a longer path, because the steep initial drop buys speed early. The first integral $y(1 + y'^2) = C$ is satisfied to 10^{-8} along the parametric solution.

PROBLEM 2 — THE BELTRAMI IDENTITY, AND ENERGY ARRIVING EARLY

Suppose $L(x, \dot{x})$ has no *explicit* dependence on t , i.e. $\partial L / \partial t = 0$. Show that along any solution of the Euler–Lagrange equation the quantity

$$E \equiv \dot{x} \frac{\partial L}{\partial \dot{x}} - L$$

is constant in time. Then evaluate it for $L = \frac{1}{2}m\dot{x}^2 - V(x)$ and say what it is.

– Solution

Differentiate E along the path, using the product rule on the first term:

$$\frac{dE}{dt} = \ddot{x} \frac{\partial L}{\partial \dot{x}} + \dot{x} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{dL}{dt}.$$

The last term is a *total* time derivative of L along the path, which by the chain rule (Chapter 0.6) is

$$\frac{dL}{dt} = \frac{\partial L}{\partial x} \dot{x} + \frac{\partial L}{\partial \dot{x}} \ddot{x} + \underbrace{\frac{\partial L}{\partial t}}_{=0}.$$

Substitute. The $\ddot{x} \partial L / \partial \dot{x}$ terms cancel between the two lines, and what is left factorises:

$$\frac{dE}{dt} = \dot{x} \left[\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) - \frac{\partial L}{\partial x} \right] = \dot{x} \cdot 0 = 0$$

by (1.2.22). So E is conserved. ■

What it is. For $L = \frac{1}{2}m\dot{x}^2 - V(x)$ we have $\partial L / \partial \dot{x} = m\dot{x}$, so

$$E = \dot{x} (m\dot{x}) - \left(\frac{1}{2}m\dot{x}^2 - V \right) = \frac{1}{2}m\dot{x}^2 + V = T + V.$$

The total energy. And notice what the hypothesis was: L *does not depend explicitly on t* — that is, the laws are the same today as yesterday. **Time-translation symmetry implies energy conservation**, and we have just proved the first instance of it without naming it.

Two forward pointers. Chapter 1.4 proves the general theorem — Noether's — of which this is one case among many: every continuous symmetry of the action yields a conserved quantity, with time translation giving energy, spatial translation giving momentum, and rotation giving angular momentum. Chapter 1.3 observes that the combination $\dot{x} \partial L / \partial \dot{x} - L$ is exactly the **Legendre transform** of L with respect to $\dot{x}$, and renames it the **Hamiltonian**. Both of those chapters are, in a real sense, this five-line calculation taken seriously.

Caution. E is the conserved quantity; it is not always the total energy. It equals $T + V$ only when the relation between generalized and Cartesian coordinates is time-independent (so that T is a homogeneous

quadratic in $\dot{q}$). Problem 3 supplies a counterexample where E is conserved and yet energy demonstrably is not.

PROBLEM 3 — BEAD ON A ROTATING HOOP

A bead of mass m slides without friction on a circular hoop of radius R , which is held in a vertical plane and driven to rotate about its vertical diameter at a fixed angular velocity Ω by a motor. Let θ be the bead's angular position measured from the bottom of the hoop. Find L and the equation of motion, identify the effective potential, and find the equilibrium positions and their stability as a function of Ω .

– Solution

Kinematics. The system has one degree of freedom: the hoop's orientation is dictated by the motor, not by the dynamics, so θ is the only coordinate. The bead's velocity has two orthogonal pieces: motion along the hoop, of magnitude $R\dot{\theta}$, and motion carried by the rotation, on a circle of radius $R \sin \theta$ about the vertical axis, of magnitude $\Omega R \sin \theta$. They are perpendicular, so

$$T = \frac{1}{2}m\left(R^2\dot{\theta}^2 + R^2\Omega^2 \sin^2 \theta\right), \quad V = -mgR \cos \theta,$$

$$L = \frac{1}{2}mR^2\dot{\theta}^2 + \frac{1}{2}mR^2\Omega^2 \sin^2 \theta + mgR \cos \theta.$$

Equation of motion. $\partial L / \partial \dot{\theta} = mR^2\dot{\theta}$ and $\partial L / \partial \theta = mR^2\Omega^2 \sin \theta \cos \theta - mgR \sin \theta$, so

$$\ddot{\theta} = \sin \theta \left(\Omega^2 \cos \theta - \frac{g}{R} \right).$$

Effective potential. The Lagrangian has the form $\frac{1}{2}mR^2\dot{\theta}^2 - V_{\text{eff}}(\theta)$ with

$$V_{\text{eff}}(\theta) = -\frac{1}{2}mR^2\Omega^2 \sin^2 \theta - mgR \cos \theta.$$

The rotation has contributed a term to T that acts, as far as the θ dynamics is concerned, exactly like a *potential* — the centrifugal potential. This is where "centrifugal force" comes from without any appeal to fictitious forces: it is a piece of kinetic energy that looks like potential energy because the coordinate it belongs to was eliminated by the motor.

Equilibria. $V'_{\text{eff}}(\theta) = mR \sin \theta (g - R\Omega^2 \cos \theta) = 0$, giving

$$\theta = 0, \quad \theta = \pi, \quad \text{and, if } \cos \theta_0 = \frac{g}{R\Omega^2} \leq 1, \quad \theta = \pm\theta_0.$$

The last pair exists only when $\Omega^2 \geq g/R$. **Stability** from the second derivative, $V''_{\text{eff}}(\theta) = -mR^2\Omega^2 \cos 2\theta + mgR \cos \theta$:

EQUILIBRIUM	V''_{eff}	STABLE?
$\theta = 0$ (bottom)	$mR(g - R\Omega^2)$	only if $\Omega^2 < g/R$
$\theta = \pi$ (top)	$-mR(g + R\Omega^2)$	never
$\theta = \pm\theta_0$	$mR^2\Omega^2 \sin^2 \theta_0$	always, where it exists

For the third row, use $mgR = mR^2\Omega^2 \cos \theta_0$ to write $V''_{\text{eff}} = -mR^2\Omega^2(2 \cos^2 \theta_0 - 1) + mR^2\Omega^2 \cos^2 \theta_0 = mR^2\Omega^2 \sin^2 \theta_0 > 0$.

What this is. As Ω is increased past the critical value $\Omega_c = \sqrt{g/R}$, the bottom of the hoop goes from stable to unstable and two new stable equilibria appear symmetrically on either side. The bead must choose one, breaking the left-right symmetry that the equations themselves possess. That is a **pitchfork bifurcation**, and it is the mechanical prototype of **spontaneous symmetry breaking**: in Chapter 6.6, V_{eff} becomes the Higgs potential, Ω becomes a parameter of the theory, and θ_0 becomes the vacuum expectation value that gives the W and Z their masses. Same picture, same algebra, different letters.

The Problem 2 caution, made concrete. L has no explicit t , so $E = \dot{\theta} \partial L / \partial \dot{\theta} - L = \frac{1}{2}mR^2\dot{\theta}^2 + V_{\text{eff}}(\theta)$ is conserved. But the true energy $T + V = \frac{1}{2}mR^2\dot{\theta}^2 + \frac{1}{2}mR^2\Omega^2 \sin^2 \theta - mgR \cos \theta$ is *not*: the two differ by $mR^2\Omega^2 \sin^2 \theta$, which changes as the bead moves. The motor is doing work on the system to hold Ω fixed. Conserved and "the energy" are different predicates.

Let $F(q, t)$ be any smooth function and define a new Lagrangian

$$L' = L + \frac{dF(q, t)}{dt} = L + \frac{\partial F}{\partial q} \dot{q} + \frac{\partial F}{\partial t}.$$

Show, in two independent ways, that L' gives exactly the same equations of motion as L : (a) by looking at what happens to S , and (b) by applying the Euler–Lagrange operator to the added term directly. Then say why this matters.

– Solution

(a) The action. The added term is a total time derivative, so by the fundamental theorem of calculus (Chapter 0.2) its integral is fixed by the endpoints alone:

$$S'[q] = \int_{t_1}^{t_2} \left(L + \frac{dF}{dt} \right) dt = S[q] + F(q(t_2), t_2) - F(q(t_1), t_1).$$

Every path we compare has the same $q(t_1)$ and $q(t_2)$ — that was the admissibility condition (1.2.4). So the two extra terms take the *same value* on every admissible path: S' and S differ by a constant on the entire space of competitors. Adding a constant to a function does not move its stationary points. Hence $\delta S' = \delta S$ and the Euler–Lagrange equations are identical. ■

(b) Directly. Write $G \equiv \frac{dF}{dt} = \frac{\partial F}{\partial q} \dot{q} + \frac{\partial F}{\partial t}$, and note that G is linear in $\dot{q}$ with coefficient $\partial F / \partial q$, so

$$\frac{\partial G}{\partial \dot{q}} = \frac{\partial F}{\partial q} \quad \implies \quad \frac{d}{dt} \frac{\partial G}{\partial \dot{q}} = \frac{d}{dt} \left(\frac{\partial F}{\partial q} \right) = \frac{\partial^2 F}{\partial q^2} \dot{q} + \frac{\partial^2 F}{\partial q \partial t}.$$

Meanwhile, differentiating G with respect to q at fixed $\dot{q}$,

$$\frac{\partial G}{\partial q} = \frac{\partial^2 F}{\partial q^2} \dot{q} + \frac{\partial^2 F}{\partial t \partial q}.$$

These are equal by the symmetry of mixed partials (Chapter 0.6), so

$$\frac{d}{dt} \frac{\partial G}{\partial \dot{q}} - \frac{\partial G}{\partial q} = 0.$$

The Euler–Lagrange operator annihilates any total time derivative, identically, for every path — not just on shell. ■ The two proofs are the same fact seen from opposite ends: (a) says the added term is invisible to the variation, (b) says it is in the kernel of the operator.

Why it matters. The map from Lagrangians to equations of motion is *many-to-one*. An entire infinite-dimensional family $\{L + \frac{dF}{dt}\}$, one member for each function $F(q, t)$, describes identical physics. That redundancy has a name: **gauge freedom**, and this is its first appearance in the book.

It is not an abstraction. ■ Quoting forward to Chapter 2.6, the Lagrangian of a particle of charge e in an electromagnetic field is $L = \frac{1}{2}m\dot{\mathbf{x}}^2 - e\varphi + e\mathbf{A} \cdot \dot{\mathbf{x}}$, where φ and $\mathbf{A}$ are the scalar and vector potentials. The fields are unchanged — and so is all measurable physics — under the substitution

$$\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi, \quad \varphi \rightarrow \varphi - \frac{\partial\chi}{\partial t}$$

for any function $\chi(\mathbf{x}, t)$. Watch what that does to L :

$$L \rightarrow L + e \left(\frac{\partial\chi}{\partial t} + \nabla\chi \cdot \dot{\mathbf{x}} \right) = L + e \frac{d\chi}{dt}.$$

Exactly a total time derivative. **The gauge invariance of electromagnetism is this problem.** Chapter 6.3 makes the same observation for a field χ that varies independently at each point of spacetime and finds that demanding it *generates* the electromagnetic field, the strong force and the weak force; Chapter 6.4 is what happens when the gauge parameter is a matrix instead of a number. All of it begins with the observation that $\frac{dF}{dt}$ integrates to a boundary term.

THE BRICK YOU JUST LAID

You can now differentiate with respect to a function. The definition (1.2.9) is Chapter 0.1's, with the displacement h promoted to an entire function η and the derivative promoted to a linear functional; nothing else about the idea changed, which is the dividend for having refused to define the derivative as a slope. From it, in four moves — differentiate under the integral, linearize, integrate by parts, apply the fundamental lemma (which you proved, with a bump function) — comes the Euler–Lagrange equation (1.2.22), one per degree of freedom.

You saw it reproduce Newton exactly, with $\partial L / \partial \dot{x}$ turning out to be the momentum and the whole equation reading $\dot{p} = F$. You saw it dispose of constraints by absorbing them into coordinates — and give them back, as multipliers, when you want the constraint force. You proved it keeps its form under *any* change of coordinates, which $F = ma$ does not, and that is the property that will let physics survive the loss of Cartesian coordinates in Chapter 3. And you saw, honestly, that $L = T - V$ is a choice justified by results, that "least" is the wrong word, and that the real reason lives in Chapter 5.8.

Where this gets spent. Chapter 1.3 Legendre-transforms this L into a Hamiltonian and discovers phase space. Chapter 1.4 acts with symmetries on this S and gets Noether's theorem — Problems 2 and the sphere grind box were both previews. Chapters 2.5 and 2.6 write the actions of a relativistic particle and of electromagnetism. Chapter 3.3 replaces the $\int ds$ of Worked Example 2 with $\int ds$ in curved spacetime and calls the answer gravity; Chapter 3.6 varies the Einstein–Hilbert action and gets the field equations. Chapter 5.2 runs §8's field version in earnest, Chapter 5.8 explains why any of it is true, and Chapters 6.4 and 7.3 write the last two lines of §8.1's table. From here to the end of the book, "solve this theory" will mean "write down its action and vary it".